1. What is a string? How can strings be created in Python?

A **string** is a sequence of characters enclosed within quotation marks. It can contain letters, numbers, spaces, and special characters.

In Python, strings can be created using:

- Single quotes - ' '
- Double quotes - " "
- Triple single quotes - ''' '''
- Triple double quotes - """ """

2. What is string indexing? Explain positive and negative indexing.

String indexing is the process of accessing an individual character of a string using its position or index.

Python uses **zero-based indexing**, meaning the first character has index 0.

Positive indexing :

Positive indexing starts from **left to right**, beginning with 0.

Character	P	Y	T	H	O	N
Positive index	0	1	2	3	4	5

Negative indexing

Negative indexing starts from **right to left**, beginning with -1.

Character	P	Y	T	H	O	N
Positive index	-6	-5	-4	-3	-2	-1

3. What is string slicing?

String slicing is used to extract a portion or substring from a string.

The general syntax is:

```
string[start:stop:step]
```

4. Write a program to find the length of a user-entered sentence.

The len() function is used to find the number of characters in a string.

Program :

```
sentence = input("Enter a sentence: ")
length = len(sentence)
print("Length of the sentence is:", length)
```

5. Write a program to count the number of vowels in a string.

We can use a for loop to examine each character

Program :

```
text = input("Enter a string: ")
count = 0
for ch in text:
    if ch.lower() in "aeiou":
    count += 1
print("Number of vowels:", count)
```

6. Write a program to reverse a string.

A simple way to reverse a string is using slicing with a step of -1.

Program :

```
text = input("Enter a string: ")
reverse = text[::-1]
print("Reversed string:", reverse)
```

7. Write a program to check whether a word is a palindrome.

A palindrome is a word or sequence that reads the same forwards and backwards.

Program :

```
word = input("Enter a word: ")
reverse = word[::-1]
if word == reverse:
    print("The word is a palindrome.")
else:
    print("The word is not a palindrome.")
```

8. Write a program to count how many times a particular character appears in a string.

Python provides the count() method to count occurrences of a character.

Program:

```
text = input("Enter a string: ")
character = input("Enter the character to count: ")
count = text.count(character)
print("The character appears", count, "times.")
```

9. Explain the difference between upper, lower, title and capitalize string operations.

Python provides several methods for changing the case of characters in a string.

1. upper()

Converts all characters to uppercase.

```
print(text.upper())
```

2. lower()

Converts all characters to lowercase.

```
print(text.lower())
```

3. title()

Converts the first letter of each word to uppercase.

```
print(text.title())
```

4. capitalize()

Converts the first character of the entire string to uppercase and makes the remaining characters lowercase.

```
print(text.capitalize())
```

10. Write a program to remove extra spaces from the beginning and end of a sentence.

The strip() method removes whitespace from the beginning and end of a string.

Program :

```
sentence = input("Enter a sentence: ")
clean_sentence = sentence.strip()
print("Sentence after removing extra spaces:")
print(clean_sentence)
```